ENGINEERING MATHEMATICS-II

T.R. Geetha
Department of Science and Humanities

ENGINEERING MATHEMATICS-II

FOURIER TRANSFORMS

T.R.Geetha
Department of Science and Humanities

PROBLEMS:

3. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1 - |t|, & if \quad |t| \le 1 \\ 0\ , & if \quad |t| > 1 \end{cases}$$

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t)\,.\,e^{-i\omega t}\,dt$$

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-1}^{1} (1 - |t|).\,e^{-i\omega}\ \ dt$$

PROBLEMS:

$$\mathcal{F}[f(t)] = \left\{ \int_{-1}^{0} 1 - (-t)\,.\,e^{-i\omega t}\,dt + \int_{0}^{1} (1-t).\,e^{-i\omega t}\,dt \right\}$$

$$\mathcal{F}[f(t)] = \int_{-1}^{0} (1+t)\,.\,e^{-i\omega t}\,dt + \int_{0}^{1} (1-t)\,.\,e^{-i\omega t}\,dt$$

$$= \left(\left[(1+t)\,\frac{e^{-i\omega t}}{-i\,\omega} \right]_{-1}^{0} - \int_{-1}^{0} 1\,.\frac{e^{-i\omega}}{-i\,\omega}\,dt \right) + \left(\left[(1-t)\,\frac{e^{-i\omega t}}{-i\,\omega} \right]_{0}^{1} - \int_{0}^{1} \left(-1\right).\frac{e^{-i\omega t}}{-i\,\omega}\,dt \right)$$

$$= \left(-\frac{1}{i\,\omega}(1-0) - \left[\frac{e^{-i}}{(-i\quad)^2} \right]_{-1}^{0} \right) + \left(-\frac{1}{i\,\omega}(0-1) + \left[\frac{e^{-iwt}}{(-i\omega)^2} \right]_{0}^{1} \right)$$

PROBLEMS:

$$\mathcal{F}[f(t)] = \left[-\frac{1}{i\,\omega} + \frac{1}{\omega^2}\left(1 - e^{iw}\right)\right] + \left[\frac{1}{i\,\omega} - \frac{1}{\omega^2}\left(e^{-i} \quad - 1\right)\right]$$

$$= \left\{\frac{1}{\omega^2} - \frac{1}{\omega^2} e^{iw} - \frac{1}{\omega^2} e^{-iw} + \frac{1}{\omega^2}\right\} = \frac{2}{\omega^2} - \frac{1}{\omega^2}\left(e^{iw} + e^{-iw}\right)$$

$$= \frac{2}{\omega^2} - \frac{2}{\omega^2} . \frac{\left(e^{iw} + e^{-iw}\right)}{2}$$

$$\mathcal{F}[f(t)] = \frac{2}{\omega^2} - \frac{2\cos\omega}{\omega^2}$$

$$\therefore \boldsymbol{\mathcal{F}[f(t)]} = \boldsymbol{\frac{2 - 2\cos\omega}{\omega^2}}$$

PROBLEMS:

4. Obtain the Fourier transform of $f(t) = \begin{cases} t^2, & |t| < a \\ 0, & |t| > a \end{cases}$, where a is positive constant.

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt$$

Since $f(t) = t^2$ for $|t| < a$ by data we have

$$F[\omega] = \int_{-a}^{a} t^2\, e^{-i\omega t}\, dt$$

PROBLEMS:

$$F[\omega] = \left[(t^2)\,\frac{e^{-i\omega t}}{-i\ \omega} - (2t)\,\frac{e^{-i\omega t}}{(-i\ \omega)^2} + (2)\,\frac{e^{-i\omega t}}{(-i\ \omega)^3}\right]_{-a}^{a}$$

$$= -\frac{1}{i\omega}\left(a^2 e^{-i\omega a} - a^2 e^{i\omega a}\right) + \frac{2}{\omega^2}\left(ae^{-i\omega a} + ae^{i\omega a}\right) + \frac{2}{i\omega^3}\left(e^{-i\omega a} - e^{i\omega a}\right)$$

$$= \frac{a^2}{i\omega}\left(e^{i\omega a} - e^{-iwa}\right) + \frac{2a}{\omega^2}\,(e^{i\omega a} + e^{-iwa}) - \frac{2}{i\ \omega^3}(e^{i\omega a} - e^{-iwa})$$

$$= \frac{a^2}{i\omega}(2\ i \sin a\ \omega) + \frac{2a}{\omega^2}(2 \cos a\omega) - \frac{2}{i\ \omega^3}(2\ i \sin a\ \omega\Big)$$

$$\left(\because \cos x = \frac{e^{ix} + e^{-ix}}{2}, \sin x = \frac{e^{ix} - e^{-ix}}{2i}\right)$$

PROBLEMS:

$$F(\omega) = \frac{2\,a^2 \sin a\,\omega}{\omega} + \frac{4\,a \cos a\,\omega}{\omega^2} - \frac{4\ \sin a\,\omega}{\omega^3}$$

$$= \frac{2a^2\omega^2 \sin a\omega + 4a\omega \cos a\omega - 4 \sin a\omega}{\omega^3}$$

$$\therefore \boldsymbol{F(\omega) = \frac{1}{\omega^3}[2\left(a^2\omega^2 - 2\right) sin\, a\omega + 4a\omega\ \ cos\, a\ \omega]}$$

THANK YOU

T.R.Geetha
Department of Science and Humanities
geethatr@pes.edu